

Week-02

13/06/2026

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Assignment No-02

Q.1. Customer rating Validation

```
→ while True:
    try:
        r = int(input("Rating: "))

        if 1 <= r <= 5:
            print("Valid rating")
        else:
            print("Rating invalid")

    except:
        print("Enter numbers only")
```

Q.2. Monthly Log Archive Manager,

```
→ try:
    f = open("logs.txt", "r")
    d = f.readlines()
    print("Lines: ", len(d))
    print("Words: ", sum(len(i.split()) for i in d))

    print("First: ", d[0].strip())
    print("Last: ", d[-1].strip())

except:
    print("File missing")
```


3) Warehouse inventory investigation,

```
→ a = [120, 85, 150, 60, 95, 180]
print("Total: ", sum(a))
print("Highest: ", max(a))
print("Lowest: ", min(a))
avg = sum(a) / len(a)
print("Average: ", avg)
print("Above 100: ",
      [i for i in a if i > 100])
```

4) Employee Attendance investigation

```
→ a = ["P", "P", "A", "P", "P", "A",
        "P", "P", "A", ""]
```

```
print("First 3: ", a[:3])
```

```
print("Last 4: ", a[-4:])
```

```
print("Alternative: ", a[::2])
```

```
print("Repeated: ", a * 3)
```


5) Fuel price revision notice

→ a = [96.5, 87.2, 95.8, 98.1, 96.3]

b = [i + 3.5 for i in a]

print("Old: ", a)

print("New: ", b)

print("Difference: ", [b[i] - a[i]

for i in
range(len(a))])

print("Average: ", sum(b) / len(b))

6) Software asset tracking dashboard.

→ import pandas as pd

d = { "Software": ["Chrome",

"VS Code", "Python",

"MySQL", "Git"],

"Count": [45, 32, 28, 20, 25]}

df = pd.DataFrame(d)

print(df)

print("Highest: ", df[df.Count ==
df.Count.max()])

print("Total: ", df.Count.sum())

print("df [df.Count > 25]")

7) Smart Factory Temperature Monitoring

→ import pandas as pd

```
a = [28.5, 30.2, 29.8, 31.1, 27.9,
      30.5, 29.0]
```

```
df = pd.DataFrame(a, column
                  = ["Temperature"])
```

```
avg = sum(a) / len(a)
```

```
print("Average: ", avg)
```

```
print("Highest: ", max(a))
```

```
print("Lowest: ", min(a))
```

```
print("Above average: ",
      [i for i in a if i > avg])
```

```
print(df)
```

8) Retail store sales analysis

→ import pandas as pd

```
d = { "product": ["Laptop",
                  "mobile", "printer", "Laptop",
                  "mobile"],
      "sales": [50000, 30000,
                 12000, 45000, 28000]}
```

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```
df = pd.DataFrame(d)
```

```
print(df)
```

```
print("Total: ", df.Sales.sum())
```

```
print("Average: ", df.Sales.mean())
```

```
print("Highest: ", df.Sales.max())
```

```
print(df[df.Sales > 25000])
```

9) Employee Salary Register.

→ import pandas as pd

```
d = { "Name": ["Aman", "Riya",  
              "Ari", "Sneha",  
              "Rahul"],  
      "Salary": [35000, 42000, 38000,  
                 45000, 40000] }
```

```
df = pd.DataFrame(d)
```

```
print(df)
```

```
print(df[df.Salary > 40000])
```

```
df["Bonus"] = df.Salary * 0.10
```

```
print(df)
```


10) Productions quality inspection

import pandas as pd

try:

f = open("quality.txt")

a = [int(i) for i in f]

df = pd.DataFrame(a,
column = ["Score"])

avg = sum(a) / len(a)

print("Average: ", avg)

print("Maximum: ", max(a))

print("Minimum: ", min(a))

print(df)

print(df[df.Score > avg])

except:

print("file missing")